

SALEM-LECKRONE, IL, USA

WMO: 724330

Lat: 38.650N

Lon: 88.967W

Elev: 570

StdP: 14.40

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 03879

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	12.4	-2.0	5.0	10.1	2.6	6.4	14.6	25.6	42.1	23.5	38.3	6.3	310	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	93.3	75.5	91.0	75.5	89.4	75.0	79.7	88.6	78.2	87.3	77.0	85.8	7.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	144.4	84.9	75.4	136.0	83.8	74.1	129.9	82.1	43.5	88.8	42.2	87.1	40.7	86.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.8	18.9	16.9	DB	0.4	96.2	5.6	2.7	-3.6	98.2	-6.9	99.8	-10.0	101.3	-14.1	103.2
			WB	-0.3	81.8	5.3	1.6	-4.1	83.0	-7.2	83.9	-10.1	84.7	-14.0	85.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.3	31.9	35.0	46.1	56.8	66.5	75.1	77.6	75.7	69.0	57.7	46.0	36.9
	DBStd	18.35	11.45	11.06	10.91	9.41	8.47	5.81	5.15	5.33	7.07	9.58	9.68	10.04
	HDD50	1891	569	434	207	41	1	0	0	0	0	31	187	421
	HDD65	4578	1026	839	591	272	91	4	1	3	37	270	572	872
	CDD50	4198	8	15	87	246	512	754	857	798	571	269	67	14
	CDD65	1410	0	0	5	28	137	308	392	336	158	44	2	0
	CDH74	13296	0	0	29	216	1073	2923	4025	3175	1462	382	11	0
	CDH80	5030	0	0	1	23	300	1137	1701	1255	509	103	1	0
Wind	WSAvg	6.7	8.8	8.5	8.6	8.5	6.7	5.5	4.3	3.9	4.1	6.1	7.5	8.0
Precipitation	PrecAvg	42.8	3.3	2.4	3.4	4.0	5.4	4.3	3.7	3.2	3.1	3.3	3.9	2.8
	PrecMax	57.2	7.8	4.5	7.7	9.6	13.7	9.1	8.7	7.2	7.8	6.9	7.8	8.3
	PrecMin	34.7	0.5	0.6	0.9	1.4	0.9	0.7	1.7	1.0	0.2	1.2	0.6	1.2
	PrecStd	6.3	1.9	1.2	1.6	2.2	3.3	2.0	1.8	1.5	1.9	1.8	1.9	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	69.1	77.9	82.5	89.7	94.8	97.2	95.8	93.4	88.4	74.8	66.2
		MCWB	59.3	60.5	62.3	65.7	72.4	74.7	75.4	75.8	73.4	71.3	61.3	59.8
	2%	DB	59.4	63.4	72.5	79.5	86.3	91.7	93.3	92.2	90.0	82.1	70.1	61.2
		MCWB	54.9	57.0	60.4	64.0	70.4	74.6	76.7	75.5	73.6	67.4	59.6	55.8
	5%	DB	54.7	57.4	67.9	76.4	83.6	89.9	90.9	89.9	86.2	78.4	65.7	56.2
		MCWB	50.1	49.8	58.1	62.6	69.6	74.2	77.1	75.2	72.3	64.6	57.4	51.9
	10%	DB	49.1	52.4	63.8	72.8	80.7	86.9	88.6	87.4	82.0	73.3	62.1	51.9
		MCWB	44.8	46.5	56.0	61.3	68.4	72.7	75.8	74.1	69.2	63.6	54.2	47.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	62.7	65.4	69.5	75.4	79.0	81.5	80.9	78.2	73.7	66.1	61.3
		MCDB	63.4	67.0	73.5	78.4	84.1	88.6	89.2	88.6	87.4	83.6	71.8	64.4
	2%	WB	56.3	57.8	62.6	67.1	73.5	77.3	80.2	79.0	75.9	71.2	62.3	57.0
		MCDB	59.1	62.8	69.6	76.4	82.3	88.2	88.9	88.1	84.7	79.1	66.9	60.4
	5%	WB	51.0	52.6	59.8	65.0	71.7	76.0	78.9	77.5	74.1	68.8	59.2	52.8
		MCDB	53.9	56.3	66.3	73.0	80.0	86.0	87.7	87.0	82.5	75.0	65.2	55.8
	10%	WB	43.9	46.1	57.0	62.7	70.0	74.6	77.7	76.0	72.3	65.1	54.9	47.9
		MCDB	49.0	51.7	63.3	70.3	77.6	83.3	85.8	84.1	79.8	71.8	61.3	51.2
Mean Daily Temperature Range	5% DB	MDBR	14.6	15.2	17.3	19.2	17.9	18.6	18.1	19.5	21.4	20.5	17.2	13.3
		MCDBR	20.4	22.4	21.8	22.8	20.5	20.9	19.9	21.9	23.9	24.3	22.6	18.5
		MCWBR	17.0	17.2	13.7	12.3	9.2	7.9	7.8	8.9	9.8	11.1	14.5	15.1
	5% WB	MCDBR	19.0	20.8	19.6	19.0	17.8	18.2	17.7	18.8	20.0	20.0	19.4	16.7
		MCWBR	17.9	18.1	13.8	12.4	9.2	8.5	8.5	9.0	9.5	10.6	15.8	15.4
Clear-Sky Solar Irradiance	taub	0.307	0.321	0.361	0.399	0.435	0.436	0.463	0.461	0.409	0.357	0.318	0.305	
	taud	2.457	2.404	2.359	2.254	2.210	2.234	2.160	2.153	2.288	2.433	2.488	2.479	
	Ebn at Noon	276	287	285	279	269	269	260	257	265	270	270	269	
	Edh at Noon	26	31	36	43	45	44	47	46	38	30	25	23	
All-Sky Solar Radiation	RadAvg	626	871	1181	1559	1752	2029	1974	1784	1492	1076	724	528	
	RadStd	64	93	110	90	153	133	128	101	102	113	79	54	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page